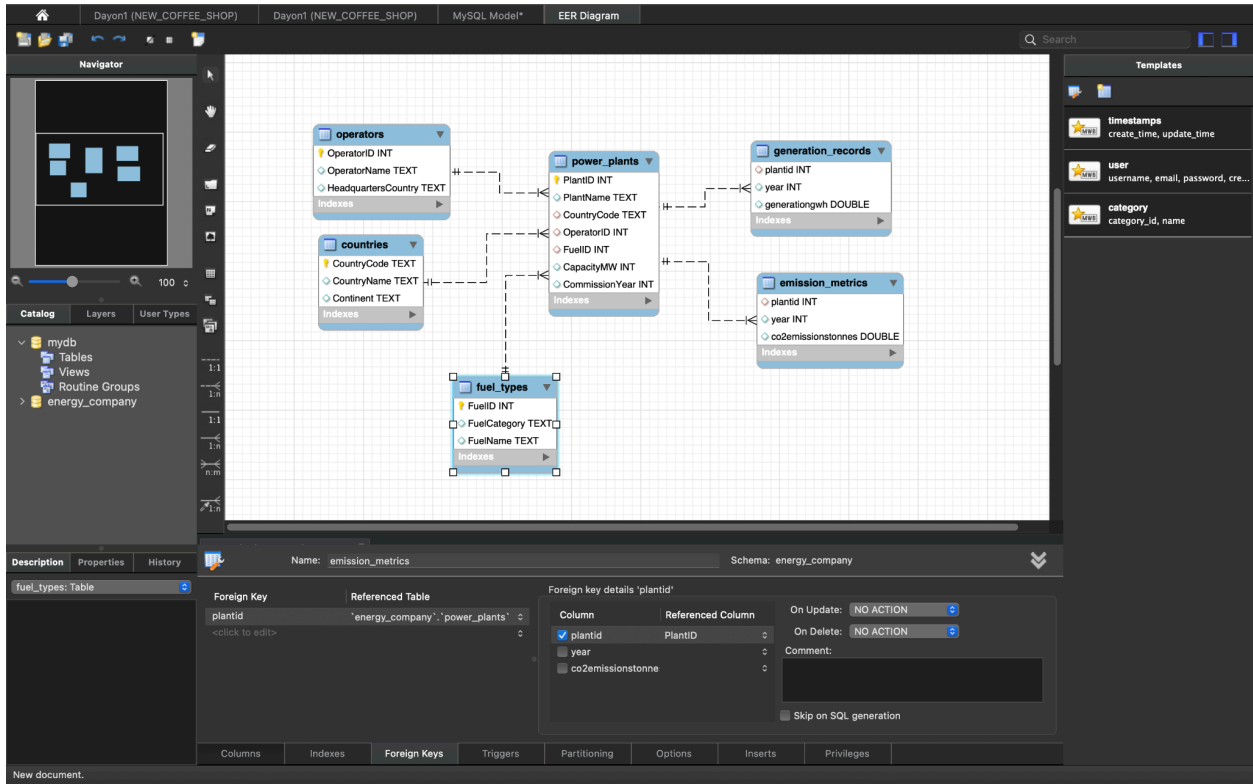




Query 1

The screenshot shows a database query tool interface. The main workspace displays the SQL query for Query 1:

```

77 PRIMARY KEY (plantid, year),
78
79 FOREIGN KEY (plantid)
80 REFERENCES power_plants(plantid)
81 );
82 -- Query 1
83
84 SELECT
85 power_plants.plantname,
86 countries.countryname,
87 operators.operatorname,
88 fuel_types.fuelcategory,
89 fuel_types.fuelname,
90 power_plants.capacitymw,
91 power_plants.commissionyear
  
```

The result grid shows the following data:

plantname	countryname	operatorname	fuelcategory	fuelname	capacitymw	commissionyear
Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
Janschwalde	Germany	Rhain Power AG	Fossil	Coal	3000	1981
Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

The bottom panel shows the query execution details:

- Query Completed
- Time: 21:17:38
- Action: SELECT
- Response: power_plants.plantname, countries.countryname, operators.o... 8 row(s) returned
- Duration / Fetch Time: 0.040 sec / 0.000023...

Query 2

Day01 (energy_company) | Day01 (NEW_COFFEE_SHOP) | MySQL Model* | EER Diagram

Administration | Schemas | SQL File 3*

SCHEMAS

energy_company

- Tables
 - countries
 - emission_metrics
 - fuel_types
 - generation_records
 - operators
 - power_plants
- Views
- Stored Procedures
- Functions

NEW_COFFEE_SHOP

- Tables
- Views
- Stored Procedures
- Functions

new_schema

sys

```
SELECT
    power_plants.plantname,
    power_plants.countrycode,
    generation_records.year,
    generation_records.generationwh
FROM power_plants
JOIN generation_records
    ON power_plants.plantid = generation_records.plantid
WHERE generation_records.year = 2024
ORDER BY generation_records.generationwh DESC;
```

108
109
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114
115
116 -- Query 3
117

100% 1:114

Result Grid

plantname	countrycode	year	generationwh
Three Gorges	CHN	2024	98800
Itaipu	BRA	2024	69000.1
Tuoketuo	CHN	2024	71000.3
Vindhyachal	IND	2024	35600.8
Grand Coulee	USA	2024	21500.5
Janschwalde	DEU	2024	18400.2
Loy Yang	AUS	2024	14200.4
Kashiwazaki-Kariwa	JPN	2024	0

Result 5

Action Output

Time	Action	Response	Duration / Fetch Time
66	21:18:31 SELECT	power_plants.plantname, power_plants.countrycode, generati... 8 row(s) returned	0.018 sec / 0.000023...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query 3

Day01 (energy_company) | Day01 (NEW_COFFEE_SHOP) | MySQL Model* | EER Diagram

Administration | Schemas | SQL File 3*

SCHEMAS

energy_company

- Tables
 - countries
 - emission_metrics
 - fuel_types
 - generation_records
 - operators
 - power_plants
- Views
- Stored Procedures
- Functions

NEW_COFFEE_SHOP

- Tables
- Views
- Stored Procedures
- Functions

new_schema

sys

```
SELECT
    power_plants.plantname,
    power_plants.countrycode,
    generation_records.year,
    generation_records.generationwh,
    emission_metrics.co2emissiontonnes
FROM power_plants
JOIN generation_records
    ON power_plants.plantid = generation_records.plantid
JOIN emission_metrics
    ON power_plants.plantid = emission_metrics.plantid
    AND generation_records.year = emission_metrics.year
WHERE generation_records.year = 2024
ORDER BY emission_metrics.co2emissiontonnes ASC;
```

122
123
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125
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100% 50:131

Result Grid

plantname	countrycode	year	generationwh	co2emissiontonn...
Grand Coulee	USA	2024	21500.5	0
Itaipu	BRA	2024	69000.1	0
Kashiwazaki-Kariwa	JPN	2024	0	0
Three Gorges	CHN	2024	98800	0
Loy Yang	AUS	2024	14200.4	1950000
Janschwalde	DEU	2024	18400.2	2450000
Vindhyachal	IND	2024	35600.8	3100000
Tuoketuo	CHN	2024	71000.3	5200000

Result 6

Action Output

Time	Action	Response	Duration / Fetch Time
67	21:18:53 SELECT	power_plants.plantname, power_plants.countrycode, generati... 8 row(s) returned	0.0083 sec / 0.00002...

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query 4

Query 4

```
135 WITH operator_generation (operatorid, total_generation) AS (  
136 SELECT  
137     power_plants.operatorid,  
138     SUM(generation_records.generationgwh)  
139 FROM power_plants  
140 JOIN generation_records  
141     ON power_plants.plantid = generation_records.plantid  
142 GROUP BY power_plants.operatorid  
143 )  
144  
145 SELECT  
146     operators.operatorname,  
147     operators.headquarterscountry,  
148     operator_generation.total_generation  
149 FROM operators  
150 JOIN operator_generation  
151     ON operators.operatorid = operator_generation.operatorid  
152 ORDER BY operator_generation.total_generation DESC;  
153
```

Result Grid

operatorname	headquarterscoun...	total_generati...
China Yangtze Power	CHN	169800.3
Electrobras	BRA	89000.1
NTPC Limited	IND	35600.8
Global Hydro Inc	USA	21500.5
Rhein Power AG	DEU	18400.2
Snowy Hydro	AUS	14200.4
Tepeco Energy	JPN	0

Action Output

Time	Action	Response	Duration / Fetch Time
68 21:19:20	WITH operator_generation (operatorid, total_generation) AS (SELECT...	7 row(s) returned	0.019 sec / 0.000026...

Query Completed

Query 5

Query 5

```
165 country_emissions (countrycode, total_emissions) AS (  
166 SELECT  
167     power_plants.countrycode,  
168     SUM(emission_metrics.co2emissionstonnes)  
169 FROM power_plants  
170 JOIN emission_metrics  
171     ON power_plants.plantid = emission_metrics.plantid  
172 GROUP BY power_plants.countrycode  
173 )  
174  
175 SELECT  
176     countries.countryname,  
177     country_generation.total_generation,  
178     country_emissions.total_emissions  
179 FROM countries  
180 JOIN country_generation  
181     ON countries.countrycode = country_generation.countrycode  
182 JOIN country_emissions  
183     ON countries.countrycode = country_emissions.countrycode  
184 ORDER BY country_generation.total_generation DESC;  
185
```

Result Grid

countryname	total_generati...	total_emissio...
China	169800.3	5200000
Brazil	89000.1	0
India	35600.8	3100000
United States	21500.5	0
Germany	18400.2	2450000
Australia	14200.4	1950000
Japan	0	0

Action Output

Time	Action	Response	Duration / Fetch Time
69 21:19:57	WITH country_generation (countrycode, total_generation) AS (SELECT...	7 row(s) returned	0.017 sec / 0.000016...

Query Completed